

From Word Counts to Dimensionality Reduction

- Distributional semantics makes intuitive sense if we think of the dimensions as representing semantic features
- For example, a *dog* is a mammal, which is terrestrial, a carnivore and often domesticated
 - A *cat* is then more similar to a dog than a goat is, since they share these traits, while goats share only some
- However, neighboring words are considerably less abstract
 - For example, *car* and *automobile* are synonyms; but are represented as distinct dimensions
 - This fails to capture similarity between a word with *car* as a neighbor and a word with *automobile* as a neighbor

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- One way to overcome this would be using dimensionality reduction methods
 - Such as singular value decomposition (Schütze, 1993) or the information bottleneck (Pereira et al., 1993)
- These are strong methods, used (with some variation) today as well
- However, we will devote the rest of the chapter to the more recent, prediction-based models

“Part of Speech Induction from Scratch”, Schütze, 1993

“Distributional clustering of English words”, Pereira, Tishby and Lee, 1993